Anqi Xu

Seattle, WA • 206-617-1059 • github.com/anqixu • linkedin.com/in/iamanqixu • anqi.xu@mail.mcgill.ca

Senior Robotics & Perception Engineer with 15+ years building field-tested autonomy and production ML systems spanning robotics, AR/VR/MR, and applied AI/ML. Experienced in building vision and ML systems that perceive, learn, and move atoms, not just bits, and in accelerating deployment speed and robustness via agentic coding workflows.

WORK EXPERIENCE

Meta (Reality Labs) — 02/2021 – Present
Senior Robotics & Computer Vision Engineer — Redmond, WA

- Improved perceptual quality KPIs, including dynamic swim from -42% to +57% prior to product launch, by architecting and building a **robotic and computer-vision test automation system**. This system tracked and isolated KPI regressions across 3 AR device programs using a nightly CI-to-dashboard pipeline.

  Tech: 3D localization, fiducial detection, world-locked rendering, sub-ms display-camera synchronization, 50-Gbps multi-camera ingestion, robot-arm control, OpenXR, C++, Python, Linux + Windows + Android, Hardware-In-The-Loop CI.
- Reduced calibration errors by up to 70% during AR prototype builds through the deployment of an **automated factory calibration-validation pipeline**. This setup evaluated camera, display, and see-through calibration stages across 5 AR/MR programs to catch factory failures before devices shipped.

  Tech: Robotic camera capture, geometric camera & display calibration, perceptual metrics (augmentation error, binocular disparity error, lateral color separation), Gage Repeatability & Reproducibility (GR&R) analysis, OpenCV, Python, Linux + Windows + Android.
- Accelerated software delivery via agentic engineering, notably condensing 6-week factory builds to 2.5 weeks. Trained the team on **AI-native infrastructure** by pioneering LLM skills to grade and debug calibration runs, and goal-validated agentic workflows for feature development, all compatible with open-source and Meta harnesses.

  Tech: Context and specification-driven engineering, agent team orchestration, across-harness skill/plugin development.
- Drove **Engineering Excellence initiatives** across the team's monorepo, including modernizing legacy data-capture framework's RPC to gRPC, and enforcing hermetic binary builds. Personally removed 2M+ lines of dead code and improved test coverage from 23% to 68%, significantly accelerating iteration speeds and reducing build times.

  Tech: gRPC and protocol buffers, hermetic across-platform builds, agent-assisted development.
- **Led a team of engineers** and multiple cross-functional partner teams to consistently ship metrology systems on schedule over 5 years. Drove org-level AR/MR product vision by translating requirements into feature roadmaps, leading design reviews, and fostering cross-discipline collaboration. Mentored interns, yielding 2 IC conversions.

Element AI — 03/2017 – 12/2020
Senior Applied Research Scientist, Robotics Program Lead — Montreal, Canada

- Led an applied-research team and built a **reinforcement-learning-based autonomous sailing agent** that uncovered novel vessel control, contributing to Team ETNZ's 2020 America's Cup victory.

  Tech: Deep Reinforcement Learning (PPO, SAC), Ray / Tune / RLlib, PyTorch, Docker, Python, AWS SageMaker, AWS S3.
- Mentored research interns and co-published an adversarial ML method for **robustness-testing SOTA visual trackers**, forcing them into losing track and locking onto inconspicuous billboards, applied to drone photography.

  Tech: domain randomization, sim-to-real transfer, visual servo, Gazebo, ROS, quadrotor control, PyTorch, OpenCV.
- Showcased rapid applied-AI solutions at NeurIPS by prototyping a system for **localizing anomalies on natural objects** (e.g., apples) utilizing instance segmentation, object-pose regression, and visual embeddings (PyTorch).

SKILLS

- Computer Vision: classical (feature detection, fiducial design, camera calibration, 3D geometry) and learning-based (object detection, segmentation, pose regression).

- Machine Learning: deep reinforcement learning, adversarial ML, probabilistic inference, attention / transformers, diffusion, foundation models (CNN, ViT, LLM/VLM).
- Robotics: localization, SLAM & VIO, planning, control; embedded software modules; electronics integration; hardware-in-the-loop; operational experience with UAVs, boats, and robot arms.
- AI-Native Engineering: context engineering, spec-driven development, skills & MCP, multi-agent orchestration.
- Research & Metrology: user-study design, robotics field experiments, statistical measurement analysis.

PUBLICATION HIGHLIGHTS

- R. Wiyatno and A. Xu. Physical Adversarial Textures that Fool Visual Object Tracking. ICCV 2019.
- A. Xu and G. Dudek. Maintaining Efficient Collaboration with Trust-Seeking Robots. IROS 2016. (Finalist, IROS KROS Best Paper Award on Cognitive Robotics)
- D. Meger *et al*. Learning Legged Swimming Gaits from Experience. ICRA 2015. (Finalist, Best Conference Paper Award)
- A. Xu and G. Dudek. OPTIMo: Online Probabilistic Trust Inference Model for Asymmetric Human-Robot Collaborations. HRI 2015.

EDUCATION

Ph.D. in Computer Science

McGill University, Canada. 2017

Thesis: Efficient Collaboration with Trust-Seeking Robots

Vanier Canadian Graduate Scholar • CIPPRS Doctoral Dissertation Award 2016

B.Eng. in Computer Engineering

McGill University, Canada. 2008

Great Distinction • Dean's Honour List

SIDE PROJECTS

- Conduit: **enabled fluid, on-the-go co-work** via an agentic assistant with modality-aware input corrections (e.g., Automatic Speech Recognition errors/typos) and adaptive output formatting (audio dialog, low-res AR displays) that orchestrates cloud and local agent harnesses (Claude Code, OpenAI Codex, Google Antigravity, OpenCode).
- Bookkeeper: **automated financial reports and outlier detection** with a personal finance system that ingests merchant, processor, and payment sources, matches transactions to reconcile a SQLite ledger, categorizes and predicts labels for card transactions using fine-tuned BERT models, and serves analytics via a FastAPI web client.
- Overcooked Bot: **built an interactive game-playing agent** via end-to-end perception, state-estimation, and control pipeline, using Faster-RCNN (Detectron2), homography-based grid detection, Kalman filtering, and PyQt annotation GUI. [target tracker, grid detection demos]
- Document Organizer: **automated the organization of scanned documents** by deploying local and cloud VLMs to categorize, rename, and file assets based on visual and semantic content.
- ueye_cam: **enabled synchronized, high-performance camera capture** for robotics research by authoring an open-source ROS driver for IDS uEye industrial cameras.

LEADERSHIP & COMMUNITY

- Program co-chair for CRV 2018 and CRV 2019.
- Reviewer for ICRA, IROS, RSS, RA-L, HRI, CoRL, CRV, ICCV, IJRR.